

PIEDMONT TRIAD, NC, USA

WMO: 723170

Lat: 36.097N

Lon: 79.943W

Elev: 271

StdP: 98.11

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 13723

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-7.5	-5.4	-18.3	0.8	-4.1	-15.8	1.0	-1.8	11.1	9.5	9.8	8.9	3.1	300	0.436

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	9.7	33.6	23.3	32.4	23.1	31.2	22.7	25.0	30.7	24.4	29.6	23.9	28.8	3.5	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
23.4	18.8	26.9	22.8	18.2	26.4	22.4	17.6	26.0	76.9	30.7	74.7	29.8	72.5	28.9	28.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
9.1	8.1	7.2	-11.5	35.4	2.9	1.4	-13.5	36.4	-15.2	37.2	-16.8	38.0	-18.9	39.0
			-12.6	25.7	2.5	0.8	-14.4	26.3	-15.9	26.7	-17.3	27.2	-19.2	27.7
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	15.3	4.1	6.1	10.1	15.3	19.7	24.0	25.7	25.0	21.6	15.7	9.8	6.0		
	DBStd	8.70	5.64	5.16	5.27	4.50	3.80	2.88	2.23	2.42	3.30	4.23	4.65	4.99		
	HDD10.0	612	198	130	66	8	0	0	0	0	0	6	59	146		
	HDD18.3	1942	442	344	259	111	32	2	0	0	10	103	256	383		
	CDD10.0	2552	14	20	68	167	301	421	488	466	348	182	55	21		
	CDD18.3	841	0	1	3	20	75	173	230	208	108	21	1	1		
	CDH23.3	7083	0	2	29	189	591	1487	2081	1789	749	156	9	1		
	CDH26.7	2544	0	0	2	32	156	560	854	681	229	29	1	0		
(14) Wind	WSAvg	3.1	3.4	3.5	3.8	3.9	3.3	3.0	2.7	2.5	2.7	2.8	2.9	3.0		
(15) Precipitation	PrecAvg	1084	83	76	96	84	94	96	110	103	103	82	77	82		
	PrecMax	1630	196	148	222	204	220	267	323	244	332	320	210	178		
	PrecMin	754	17	12	17	10	14	0	27	18	0	0	6	15		
	PrecStd	186	40	33	43	44	48	58	63	57	75	60	50	40		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	20.6	22.8	26.4	29.7	31.7	34.5	35.1	35.0	32.9	29.7	24.7	22.0		
		MCWB	14.9	15.8	16.1	18.3	21.4	22.8	24.1	23.6	21.8	21.9	17.0	16.1		
	2%	DB	17.8	19.6	23.7	27.4	29.9	32.8	33.6	33.2	31.0	27.0	21.9	18.9		
		MCWB	14.2	13.6	14.9	17.6	20.8	22.7	23.8	23.5	22.0	19.5	15.4	15.3		
	5%	DB	15.7	17.4	21.4	25.6	28.5	31.3	32.3	31.8	29.4	24.9	20.0	16.6		
		MCWB	12.2	12.5	13.7	16.8	20.2	22.4	23.5	23.2	21.5	18.3	15.2	13.4		
	10%	DB	13.0	14.7	18.8	23.6	26.8	29.9	31.1	30.3	27.7	23.0	17.9	14.2		
		MCWB	9.6	9.7	12.5	15.9	19.3	21.8	23.1	22.8	20.8	17.8	13.9	10.9		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	17.4	17.7	18.0	20.8	23.1	24.6	25.7	25.7	24.5	23.5	19.8	18.6		
		MCDB	18.8	20.3	22.9	25.2	28.4	30.7	32.2	31.4	29.2	26.3	21.3	20.5		
	2%	WB	15.2	15.6	16.6	19.4	22.1	23.9	25.1	25.0	23.6	21.7	18.0	16.2		
		MCDB	17.3	18.5	21.0	24.8	27.8	29.8	31.1	30.3	27.8	24.6	20.1	18.1		
	5%	WB	12.9	13.3	15.4	18.2	21.3	23.4	24.5	24.3	22.8	20.2	16.4	14.2		
		MCDB	15.0	16.3	19.5	23.4	26.4	29.0	30.1	29.1	26.4	23.3	18.9	16.1		
	10%	WB	10.1	10.7	13.9	17.1	20.5	22.8	23.9	23.7	22.1	18.7	14.5	11.4		
		MCDB	12.6	13.5	17.8	21.8	25.1	28.1	29.1	28.2	25.4	22.0	17.4	13.5		
(35) Mean Daily Temperature Range	5% DB	MDBR	9.8	10.5	11.2	11.7	10.8	10.0	9.7	9.5	9.6	10.9	11.1	9.5		
		MCDBR	12.3	13.7	13.8	13.9	12.4	11.5	11.2	11.1	11.7	12.7	13.3	11.8		
	5% WB	MCWBR	8.7	9.1	7.5	6.3	5.1	4.1	3.7	3.6	4.4	6.2	8.2	8.6		
		MCWBR	10.9	11.9	11.7	11.6	10.5	10.3	10.0	9.4	9.2	9.8	10.5	10.0		
(40) Clear-Sky Solar Irradiance	taub		0.304	0.318	0.349	0.391	0.434	0.473	0.515	0.506	0.410	0.368	0.320	0.306		
	taud		2.513	2.468	2.422	2.306	2.216	2.144	2.050	2.057	2.323	2.420	2.513	2.540		
	Ebn at Noon		899	923	920	892	855	818	780	778	844	853	871	873		
	Edh at Noon		80	95	108	128	143	153	167	163	119	98	80	74		
(44) All-Sky Solar Radiation	RadAvg		2.53	3.20	4.25	5.32	5.88	6.32	5.89	5.39	4.50	3.68	2.79	2.17		
	RadStd		0.17	0.39	0.29	0.42	0.48	0.46	0.39	0.27	0.41	0.47	0.22	0.19		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		+0.61	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+181	+116
(47) Regional (62 neighbors)		+0.43	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+140	+96

Nomenclature: See separate page